Roll No. :.....

Total Printed Pages: 02

EIA/ECA-101
Examination - June- 2023
B.Tech. I Sem : EI, EC
Basic Electronics

Max. Marks : 60
Min. Marks : 19

Time : 2 Hrs

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) List the applications of LED. 02

(b) Name the types of breakdowns for a Zener Diode. 02

(c) Differentiate p type and n type semiconductor material. 03

OR

Draw the VI characteristics of PN junction diode. 03

(d) With the help of VI Characteristics explain the working of tunnel diode. 05

OR

With the help of VI Characteristics explain the working of Zener diode. 05

Q.2 (a) Why emitter is heavily doped? 02

(b) What are the three types of transistor configurations? 02

(c) Derive the relation between alpha and beta of transistor. 03

OR

List different biasing techniques of BJT. 03

(d) Compare CB, CE, CC configuration. 05

OR

Sketch the Circuit diagram of a NPN common base configuration explain its working and VI characteristics. 05

Q.3 (a) What is the effect of temperature change on h parameters of a transistor? 02

(b) Write short note on the four h-parameter. 02

(c) Why do we need hybrid parameters? 03

OR

Draw the hybrid equivalent circuit for common emitter transistor. 03

(d) Derive the equations for current gain and voltage gain of a CE amplifier using h-parameters. 05

OR

Write about Small Signal low frequency h-parameter model of BJT and explain the same. 05

Q.4 (a) List different types of coupling techniques. 02

(b) State cascade amplifier. 02

(c) What are the advantages of Representation of Gain in Decibels? 03

OR

A multistage amplifier employs five stages each of which has a power gain of 30. What is the total gain of the amplifier in db? 03

(d) Draw the circuit and frequency response of RC coupled amplifier and explain. 05

OR

Write the operation of RC Coupled amplifier 05

Q.5 (a) What do you mean by tuned amplifiers? 02

(b) Mention any two Applications of tuned amplifiers. 02

(c) Compare single tuned and double tuned amplifiers. 03

OR

Discuss harmonic Distortion. 03

(d) Discuss the characteristics and response of double tuned amplifiers. 05

OR

Differentiate A, B, AB and C type of power amplifier. 05

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